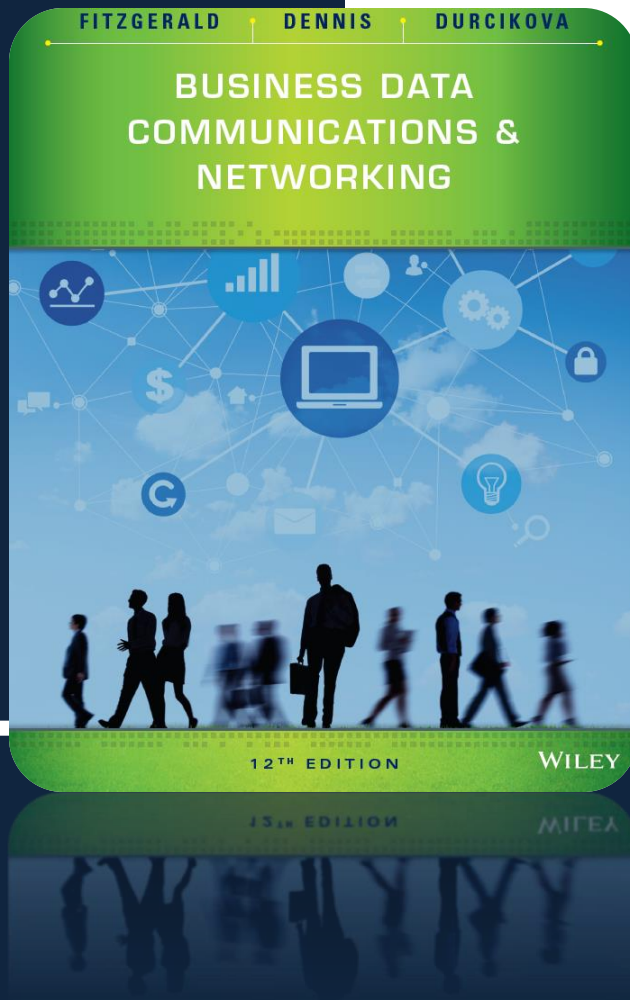




BUSINESS DATA COMMUNICATIONS & NETWORKING

Chapter 4 Data Link Layer

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Outline

- Data Link Layer
 - Media Access Control
 - Controlled Access
 - Contention Access
 - Error Control
 - Prevention
 - Detection
 - Correction
 - Protocols
 - Async
 - SDLC
 - Ethernet
 - PPP
 - Transmission Efficiency and Throughput
- Implications for Management

Data Link Layer

- Layer 2 in the Internet model
- Responsible for moving messages from one device to another
- Controls the way messages are sent on media
- Organizes physical layer bit streams into coherent messages for the network layer
- Major functions of a data link layer protocol
 - Media Access Control
 - Error Control
 - Message Delineation (description)

Internet Model

Application

Transport

Network

Data Link

Physical

Outline

- Media Access Control
 - Controlled Access
 - Contention Access
- Error Control
 - Prevent
 - Detect
 - Correct
- Protocols
 - Message Delineation
 - Ethernet
 - PPP
 - Transmission Efficiency

Media Access Control

- Controls **which device** transmits and **when**
- Important on
 - Multipoint (shared) circuits
 - Half-duplex point-to-point circuits
- Two approaches to control
 1. Contention access
 2. Controlled access

Media Access Control

- Contention
 - Transmit **whenever circuit is available** with **no** centralized **control**
 - Common in Ethernet LANs
 - When devices transmit at the same time, a **collision** occurs
 - Devices must be “polite” and follow these steps:
 1. **“Listen” for traffic**
 2. **If another device is transmitting, wait to transmit**
 3. **Otherwise, transmit (and keep listening)**
 4. **If another device begins to transmit, stop and wait**

Media Access Control

- Controlled access
 - Common in wireless LANs
 - **Access Request**
 - Each device must **get “permission”** to transmit, similar to raising a hand
 - **Polling**
 - **Roll-call polling**
 - Central device (**controller**) determines which devices can transmit
 - Each client is checked **periodically** to see if it needs to transmit
 - **Hub Polling (token passing)**
 - One device begins the poll and then passes it to another device until it reaches them all

Media Access Control

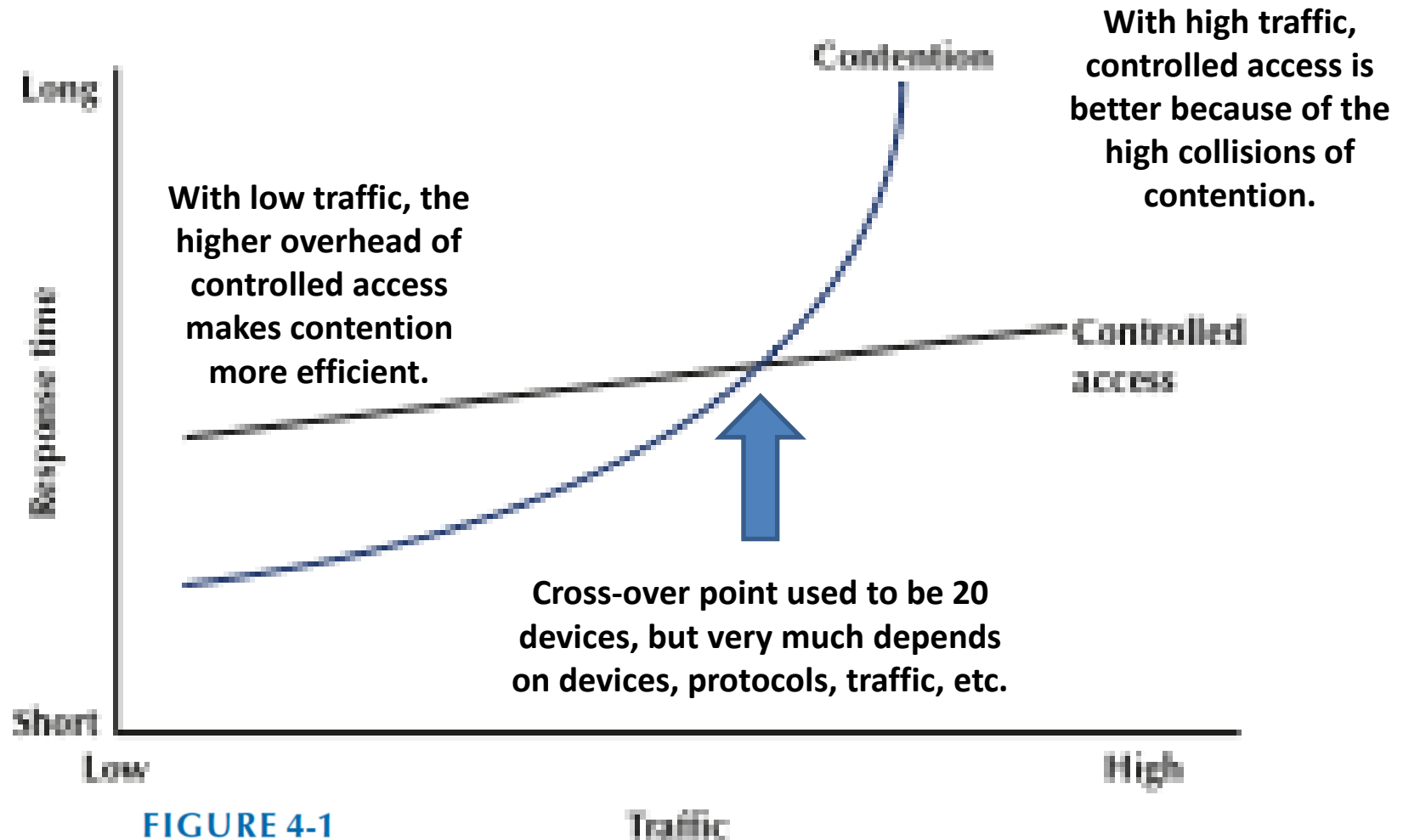


FIGURE 4-1
Relative response times

Error Control

- Network errors
 - Types
 - Corrupted data
 - Lost data
 - Caused by problems in transmission (not humans)
- Networks should be designed with:
 - Error prevention
 - Error detection
 - Error correction

Sources of Network Errors

- Line noise and distortion
 - Major reason for errors and caused by several sources
 - More likely on electrical media
 - Undesirable electrical signal
 - Degrades performance of a circuit
 - Manifestation (appearance)
 - Extra bits
 - “Flipped” bits
 - Missing bits

Error Prevention

Source of Error	What Causes It	How to Prevent or Fix
White Noise	Movement of electrons	Increase signal strength
Impulse Noise	Sudden increases in electricity (e.g., lightning)	Shield or move the wires
Cross-talk	Multiplexer guardbands too small or wires too close together	Increase the guardbands or move or shield the wires
Echo	Poor (misaligned) connections	Fix the connections or tune equipment
Attenuation	Gradual decrease in signal over distance	Use repeaters
Intermodulation noise	Signals from several circuits combine	Move or shield the wires

Error Detection

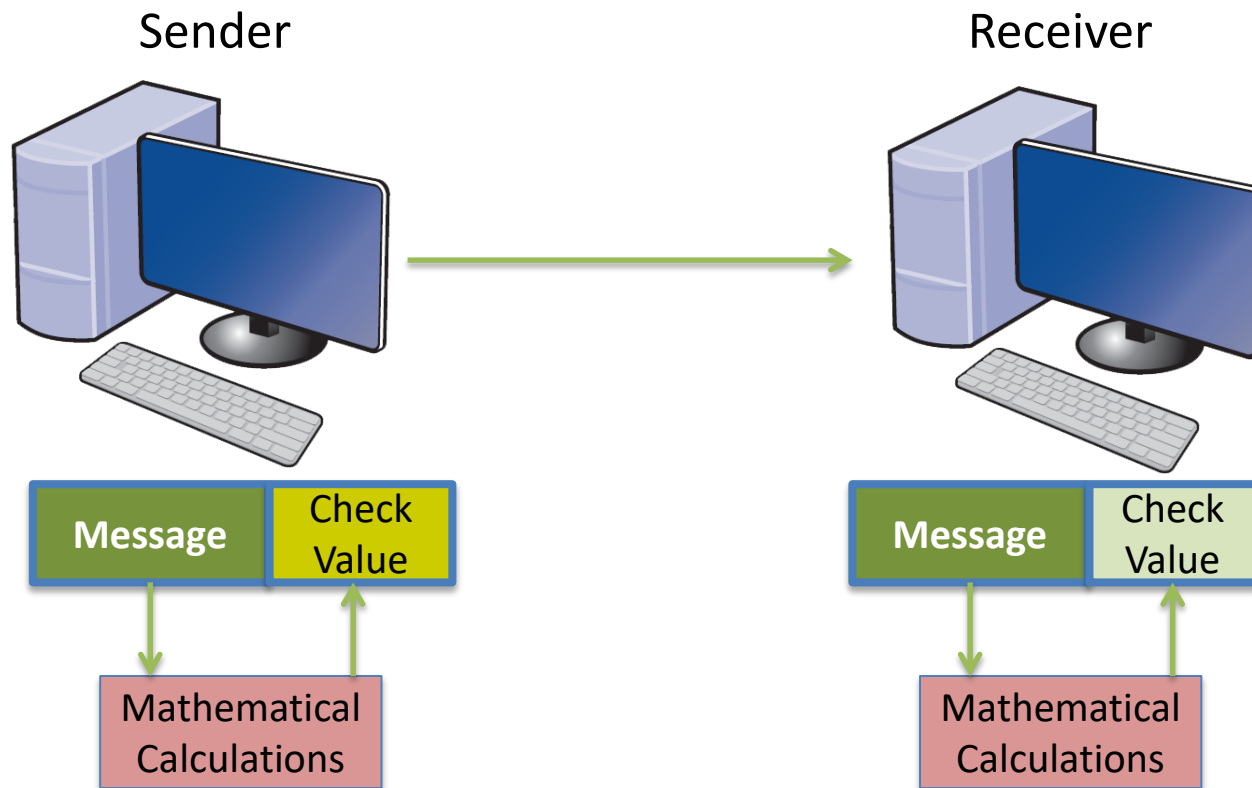
- Receivers need to know when the data transmitted is not correct
- Add “**check value**” (error detection value) to message



- Check value produced by **mathematical formula**

Error Detection

- Both sender and receiver calculate check value
- Sender tests whether the check values match



Error Detection

- Parity check
 - 1-bit check value
 - Based on the number of 1's in the message
 - Even parity: number of 1's remains even
 - Odd parity: number of 1's remains odd
 - Simple, but only detects 50% of errors

Example (Even Parity)

Character: 'A'
Binary: 01000001
Parity Bit: 0

Sum = 2 (even)



Character: 'C'
Binary: 01000011
Parity Bit: 1

Sum = 3 (odd)



Error Detection

- Checksum
 - 1-byte (typically) check value
 - Checksum algorithms vary in creation of check values
 - Detects 95% of errors
- Example checksum algorithm:
 - Add the decimal value of each character in the message
 - Divide the sum by 255
 - Use the remainder as the checksum

Error Detection

- Cyclic redundancy check (CRC)
 - Treats message as a single binary number
 - Divides by a preset number
 - Uses **remainder** as the check value
- Preset number is chosen so that remainder is the correct number of bits
- Modes:
 - CRC-16 (16 bits added to the message, ~**99.998%** error detection rate)
 - CRC-32 (32 bits added to the message, >**99.99999%** error detection rate)

Error Correction

- Once detected, errors must be corrected
- Error correction techniques
 - Retransmission (or backward error correction)
 - Retransmission is simple and most common
 - Automatic Repeat reQuest (ARQ)
 - Stop-and-wait ARQ
 - Continuous ARQ
 - This can also provide flow control by limiting the number of messages sent
 - Forward error correction
 - Receiving device can correct messages without retransmission

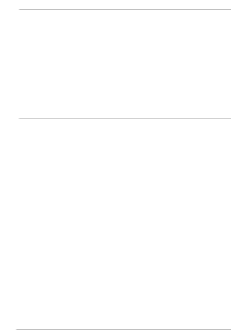
Error Correction

- Stop-and-wait ARQ

Sender

Receiver

1. Receiver receives frame and sends:
 - Acknowledgement (ACK) if no error
 - Negative acknowledgement (NAK) if error
2. If NAK, sender re-sends data



Error Correction

- Stop-and-wait ARQ
 3. If no ACK or NAK, Sender retransmits frame after “timeout”
 4. If no ACK, sender re-sends data, receiver sends ACK and deletes duplicate frame

Sender

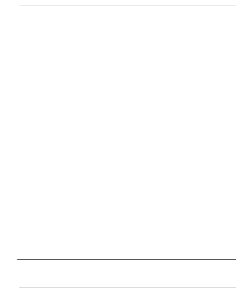
Receiver

Error Correction

Sender

Receiver

- Continuous ARQ
 - Sender does not wait for ACKs before sending more data
 - “Sliding window” is number of frames allowed to be unacknowledged by receiver
 - Agreed upon by sender and receiver



Error Correction

- Forward error correction
 - Includes a certain level of **redundancy in transmitted data** so that receiving device can correct errors
(Ranges from a small percentage of extra bits to 100 % redundancy where number of error detecting bits equals number of data bits.)
 - **Does not require retransmission**
 - Used **only when retransmission is impossible, very costly, or time consuming** (e.g., satellite connections)

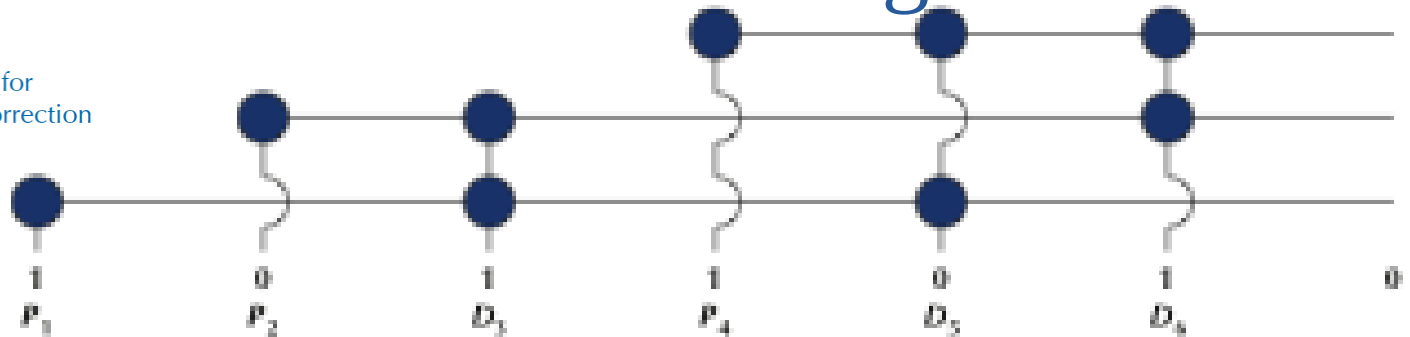
A trip from the Earth station to the satellite and back includes a significant delay.

Some weather conditions make it impossible to transmit without some errors,

Compared to satellite equipment costs, additional cost of forward error correction is insignificant.

Error Correction - Hamming Code

FIGURE 4-6
Hamming code for
forward error correction



Checking relations between parity bits (P) and data bits (D)

Each data bit
figures into three
EVEN parity bit
calculations

If any one bit
(parity or data)
changes →
change in data
bit can be
detected and
corrected

$\checkmark$ = Corresponding parity check is correct $\times$ = Corresponding parity check fails			Determines in which bit the error occurred	
P_4	P_2	P_1		
$\checkmark$	$\checkmark$	$\checkmark$	→	no error
$\checkmark$	$\checkmark$	$\times$	→	P_1
$\checkmark$	$\times$	$\checkmark$	→	P_2
$\checkmark$	$\times$	$\times$	→	D_3
$\times$	$\checkmark$	$\checkmark$	→	P_4
$\times$	$\checkmark$	$\times$	→	D_5
$\times$	$\times$	$\checkmark$	→	D_6
$\times$	$\times$	$\times$	→	D_7

Only
works for
one bit
errors

Interpreting parity bit patterns

Error Correction

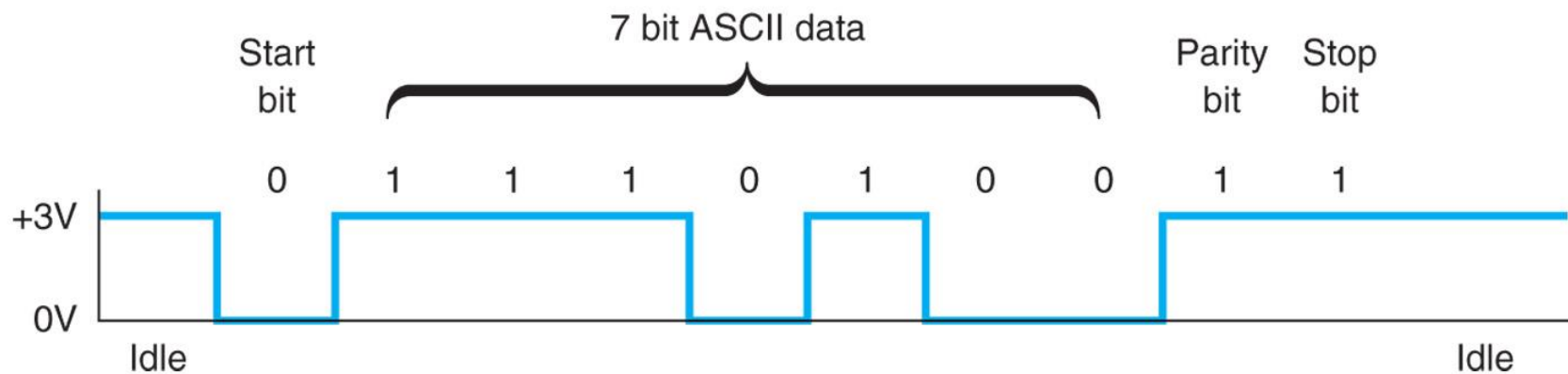
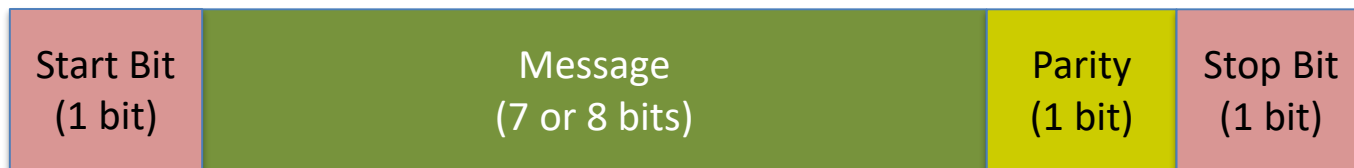
- Error control in practice
 - On **wired** connections, errors are quite **rare**
 - Most data link layer software today does not correct errors, **only detect** them and **discard frames with errors**
 - Error correction must then be done at a higher layer (**Transport layer**)

Data Link Protocols

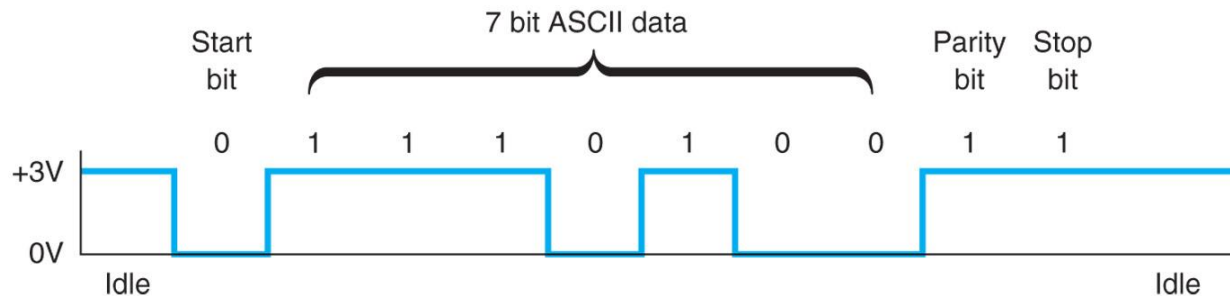
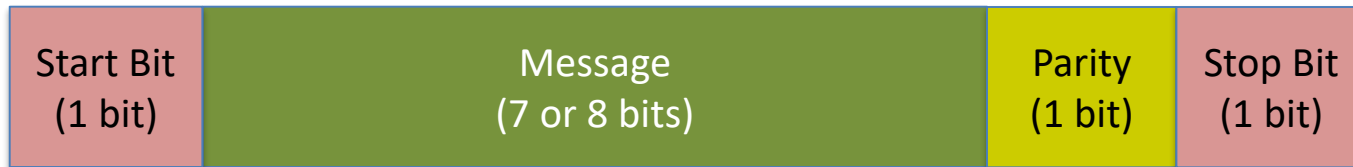
- Classification
 - Asynchronous transmission
 - Synchronous transmission
- Protocols differ by
 - Message delineation
 - Frame length
 - Frame field structure

Data Link Protocols

- Asynchronous serial transmission (async)
 - **Old** protocol (e.g., used in teletype)
 - Transmits **one character at a time**
 - Delineation indicated by **start and stop bits**



Data Link Protocols



Show how the word “HI” would be sent using asynchronous transmission using even parity (make assumptions about the bit patterns needed).

H → 0100 1000

I → 0100 1001

Adding single bit, even parity:

H → 0100 1000 0

I → 0100 1001 1

Thus, asynchronous with 0 start bit, 1 stop bit:

0 0100 1000 0 1

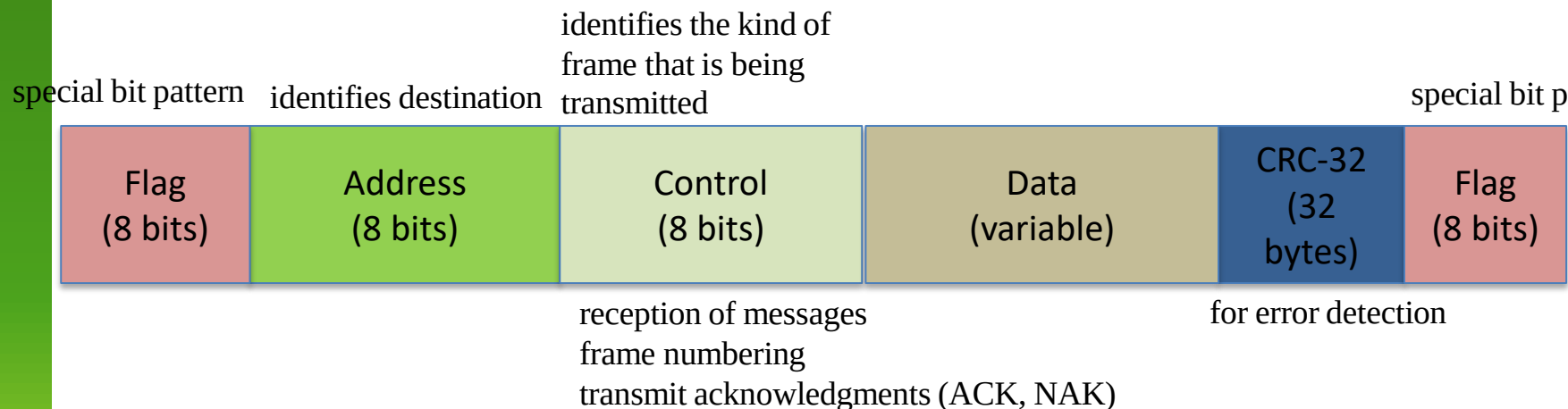
0 0100 1001 1 1

Data Link Protocols

- Synchronous transmission
 - Data sent in a **large block called a frame**
 - Includes **addressing information**
 - Includes **synchronization characters** to let the receiver know **when data transmission begins**
 - Example protocols: SDLC, HDLC, Ethernet, PPP

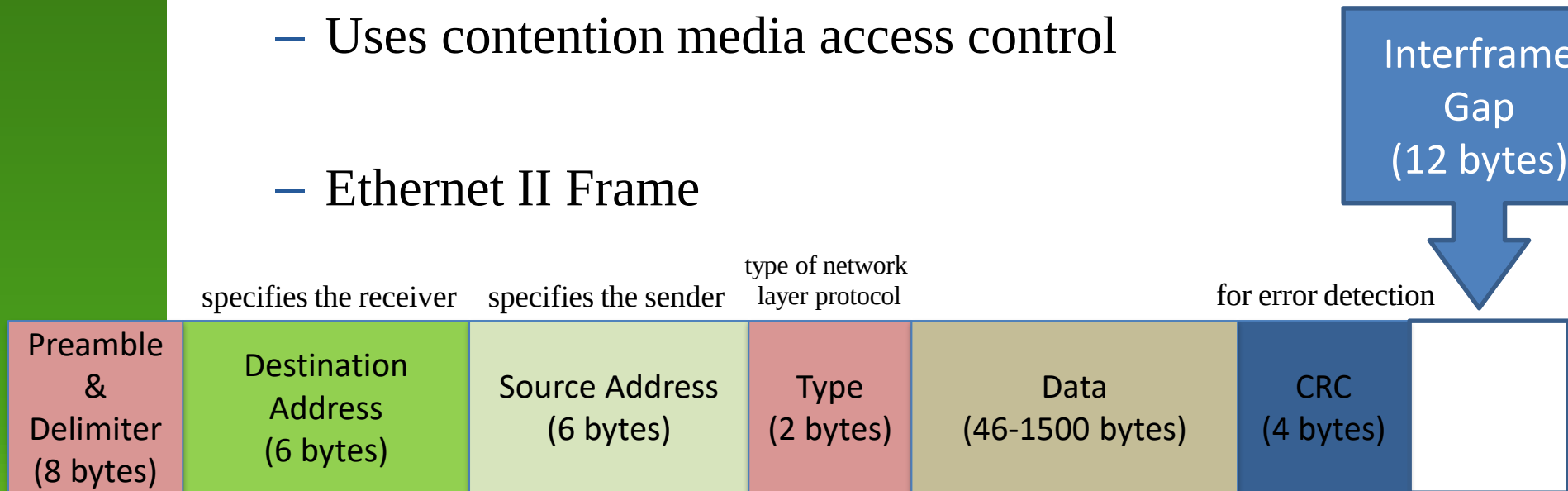
Data Link Protocols

- Synchronous Data Link Control (SDLC)
 - Synchronous bit-oriented protocol developed by IBM
 - Uses bit stuffing (zero insertion) to overcome transparency problem



Data Link Protocols

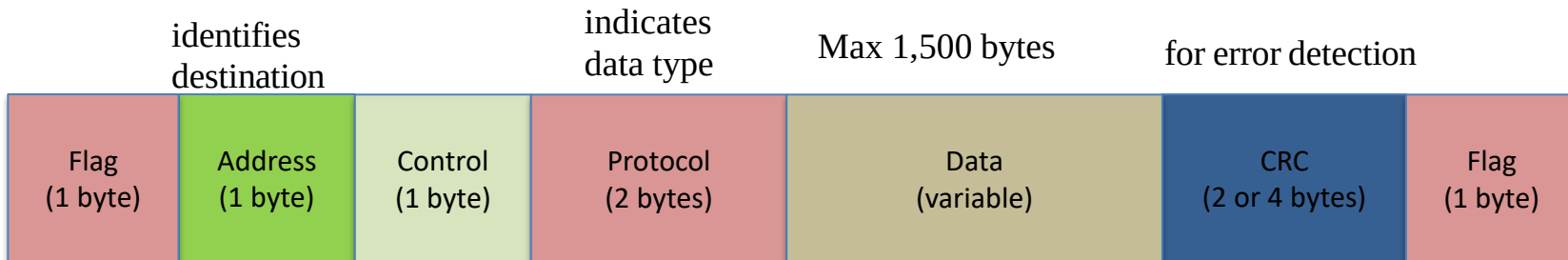
- Ethernet
 - IEEE 802.3 standard and Ethernet II
 - Most widely used LAN protocol
 - Uses contention media access control
 - Ethernet II Frame



38 Total Bytes of Overhead

Data Link Protocols

- Point-to-point protocol (PPP)
 - Common WAN protocol
 - Similar to the SDLC frame



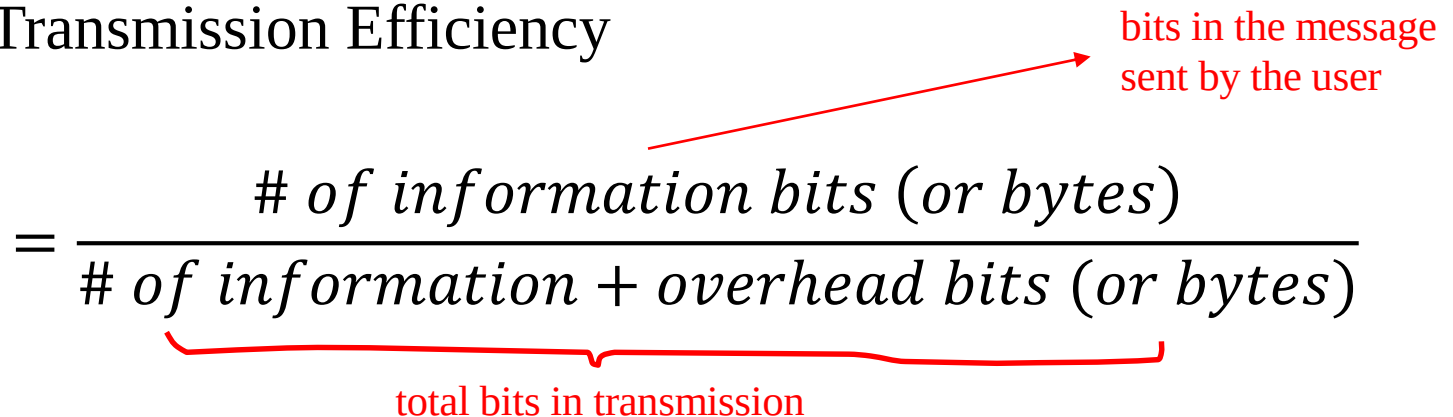
reception of messages
frame numbering
transmit acknowledgments (ACK, NAK)

Transmission Efficiency

- Protocol design contributes to network efficiency
- Transmission Efficiency

$$= \frac{\text{\# of information bits (or bytes)}}{\underbrace{\text{\# of information + overhead bits (or bytes)}}_{\text{total bits in transmission}}}$$

bits in the message
sent by the user



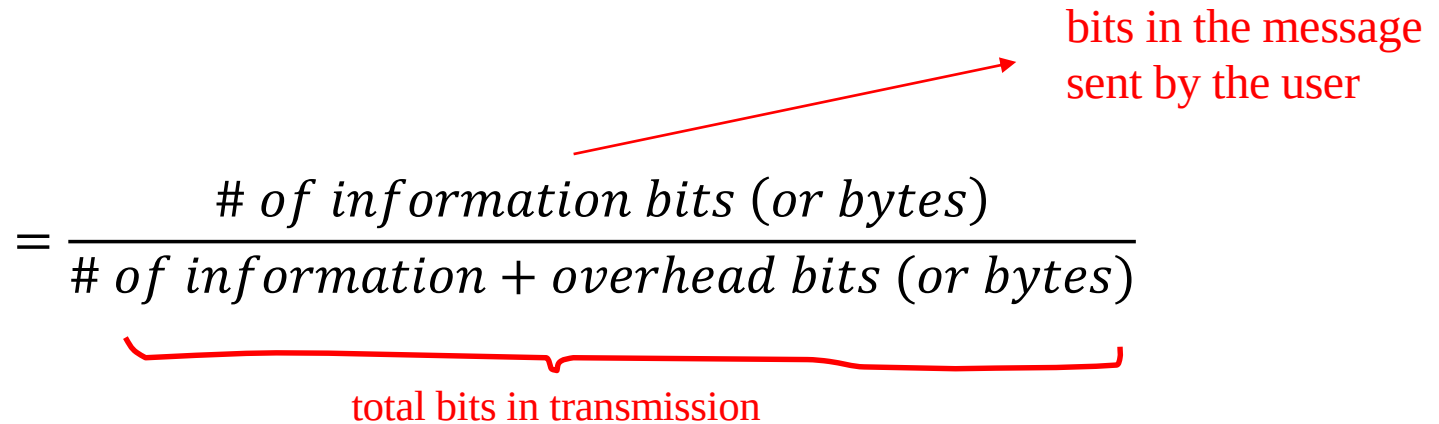
Information bits are used to convey the **user's meaning**.

Overhead bits are used for **error checking, and marking the start and end of characters and packets**.

Transmission Efficiency

$$= \frac{\text{\# of information bits (or bytes)}}{\underbrace{\text{\# of information + overhead bits (or bytes)}}_{\text{total bits in transmission}}}$$

bits in the message
sent by the user

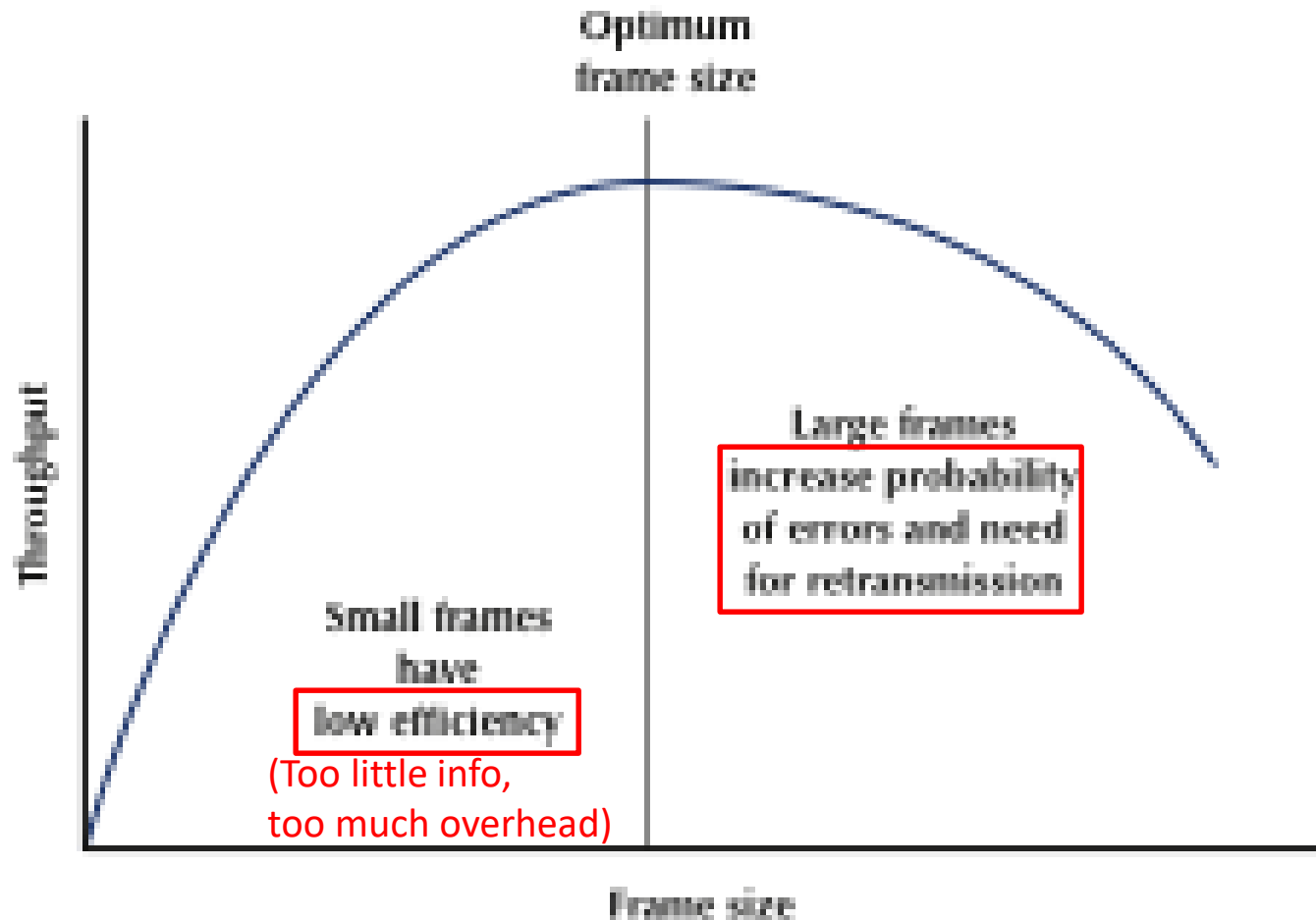


- Async Efficiency = $8/(8+3) = 0.73 = 73\%$
- How can we increase efficiency?
 - Increase Data bits
 - Decrease Overhead bits
- What is the efficiency if you send 342 bytes over Ethernet II? $342/(342+38)$
- What is the efficiency if you send 722 bytes over Ethernet II? $722/(722+38)$

Data Link Protocols

FIGURE 4-12

Frame size effects on throughput

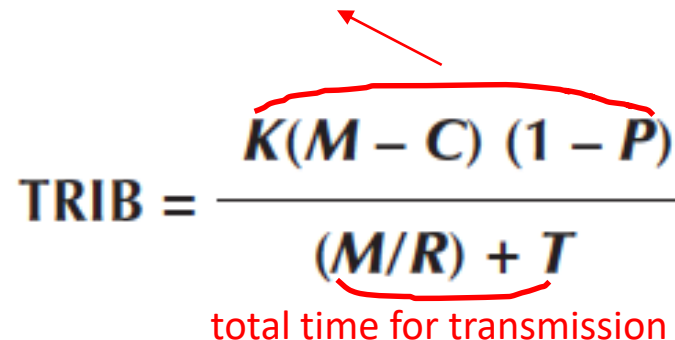


Throughput

- More complicated than simple protocol efficiency because it depends on the **retransmission rate, transmission rate, delay**
- **Transmission Rate of Information Bits (TRIB)** is a measure of **effective number of bits transmitted in unit of time**

$$\text{TRIB} = \frac{K(M - C)(1 - P)}{(M/R) + T}$$

total time for transmission



K = information bits per character

M = frame length in characters

R = data transmission rate in characters/sec

C = average number of non-information characters per frame (control characters)

P = probability that a frame will require retransmission

T = time delay between frames (in seconds)

Implications for Management

- Industry has seen a rapid consolidation in the number of data link layer protocols
- Use fewer, well-established protocols for simpler network management, greater access to trained experts, and less expensive equipment